

Ch23: 2 , 4 , 9 , 24 , 33 , 47 , 49

Basic score 4` + problem 4 2`+ problem 24 2`+ problem 49 2`

2. The field due to the uniform distributed large sheet is $E = \frac{\sigma}{2\epsilon_0}$. The force (in magnitude) on the electron (due to that field) is $F = eE$, and assuming it's the *only* force then the acceleration is

$$a = \frac{F}{m} = \frac{eE}{m} = \frac{e\sigma}{2\epsilon_0 m} = \text{slope of the graph} (= 1.6 \times 10^5 \text{ m/s divided by } 7.0 \times 10^{-12} \text{ s}).$$

Thus we obtain $\sigma = 2.3 \times 10^{-6} \text{ C/m}^2$.

4. (a) Since the charge distribution has cylindrical symmetry, so we can choose a coaxial cylindrical shell with radius r and height L as the Gaussian surface. Then by applying Gauss law we have:

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} \Rightarrow E \cdot 2\pi r L = \frac{\lambda L}{\epsilon_0}$$

$$\Rightarrow \vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{(1.5 \times 10^{-9} \text{ C/m})}{2\pi\epsilon_0 (0.15 \text{ m})} = 1.8 \times 10^2 \text{ N/C.}$$

公式 0.5 point+结果 0.5 point

(b) Since the field is zero inside the conductor (in an electrostatic configuration), we choose a coaxial cylindrical shell with radius r_{in} and height L as the Gaussian surface. Applying Gauss law, we have:

$$\left. \begin{aligned} \oint \vec{E} \cdot d\vec{A} &= \frac{q_{enc}}{\epsilon_0} \\ E &= 0 \end{aligned} \right\} \Rightarrow q_{enc} = 0 = q_{line} + q_{in} = \lambda L + \sigma_{inner} 2\pi r_{in} L$$

$$\Rightarrow \sigma_{inner} = -\frac{\lambda}{2\pi r_{in}} = -\frac{1.5 \times 10^{-9} \text{ C/m}}{2\pi (0.050 \text{ m})} = -4.8 \times 10^{-9} \text{ C/m}^2$$

公式 0.3point+结果 0.2point

for the inner surface.

(c) The total charge is conserved for the cylindrical conducting shell, so we have:

$$q_{in} + q_{out} = 0 \Rightarrow \sigma_{inner} 2\pi r_{in} L + \sigma_{outer} 2\pi r_{out} L$$

$$\Rightarrow \sigma_{out} = -\frac{r_{in}}{r_{out}} \sigma_{inner} = -\frac{5\text{cm}}{10\text{cm}} \sigma_{inner} = -\frac{1}{2} \sigma_{inner} = 2.4 \times 10^{-9} \text{ C/m}^2$$

公式 0.3point+结果 0.2point

9. To find an expression for the electric field inside the shell in terms of A and the distance from the center of the shell, select A so the field does not depend on the distance. We use a Gaussian surface in the form of a sphere with radius r_g , concentric with the spherical shell and within it ($a < r_g < b$). Gauss' law will be used to find the magnitude of the electric field a distance r_g from the shell center. The charge that is both in the shell and within the Gaussian sphere is given by the integral $q_s = \int \rho dV$ over the portion of the shell within the Gaussian surface. Since the charge distribution has spherical symmetry, we may take dV to be the volume of a spherical shell with radius r and infinitesimal thickness dr : $dV = 4\pi r^2 dr$. Thus,

$$q_s = 4\pi \int_a^{r_g} \rho r^2 dr = 4\pi \int_a^{r_g} \frac{A}{r} r^2 dr = 4\pi A \int_a^{r_g} r dr = 2\pi A (r_g^2 - a^2).$$

The total charge inside the Gaussian surface is

$$q + q_s = q + 2\pi A (r_g^2 - a^2).$$

The electric field is radial, so the flux through the Gaussian surface is $\Phi = 4\pi r_g^2 E$, where E is the magnitude of the field. Gauss' law yields

$$4\pi \epsilon_0 E r_g^2 = q + 2\pi A (r_g^2 - a^2).$$

We solve for E :

$$E = \frac{1}{4\pi \epsilon_0} \left[\frac{q}{r_g^2} + 2\pi A - \frac{2\pi A a^2}{r_g^2} \right].$$

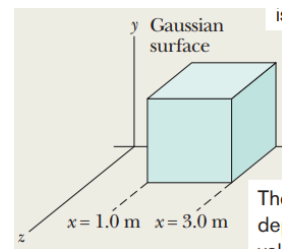
For the field to be uniform, the first and last terms in the brackets must cancel. They do if $q - 2\pi A a^2 = 0$ or $A = q/2\pi a^2$. With $a = 1.5 \times 10^{-2} \text{ m}$ and $q = 75.0 \times 10^{-15} \text{ C}$, we have $A = 5.3 \times 10^{-11} \text{ C/m}^2$.

24. We use $\Phi = \int \vec{E} \cdot d\vec{A}$ and note that the side length of the cube is $(3.0 \text{ m} - 1.0 \text{ m}) = 2.0 \text{ m}$.

(a) On the top face of the cube $y = 2.0 \text{ m}$ and $d\vec{A} = (dA)\hat{j}$. Therefore, we have

$$\vec{E} = 5\hat{i} - 3((2.0)^2 + 2)\hat{j} = 5\hat{i} - 18\hat{j}. \text{ Thus the flux is}$$

$$\Phi = \int_{\text{top}} \vec{E} \cdot d\vec{A} = \int_{\text{top}} (5\hat{i} - 18\hat{j}) \cdot (dA)\hat{j} = -18 \int_{\text{top}} dA = (-18)(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = -72 \text{ N} \cdot \text{m}^2/\text{C}.$$



过程 0.3point+计算结果 0.2point

(b) On the bottom face of the cube $y = 0$ and $d\vec{A} = (dA)(-\hat{j})$. Therefore, we have

$E = 5\hat{i} - 3(0^2 + 2)\hat{j} = 5\hat{i} - 6\hat{j}$. Thus, the flux is

$$\Phi = \int_{\text{bottom}} \vec{E} \cdot d\vec{A} = \int_{\text{bottom}} (5\hat{i} - 6\hat{j}) \cdot (dA)(-\hat{j}) = 6 \int_{\text{bottom}} dA = 6(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = +24 \text{ N} \cdot \text{m}^2/\text{C}.$$

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(c) On the left face of the cube $d\vec{A} = (dA)(-\hat{i})$. So

$$\Phi = \int_{\text{left}} \vec{E} \cdot d\vec{A} = \int_{\text{left}} (5\hat{i} + E_y\hat{j}) \cdot (dA)(-\hat{i}) = -5 \int_{\text{bottom}} dA = -5(2.0)^2 \text{ N} \cdot \text{m}^2/\text{C} = -20 \text{ N} \cdot \text{m}^2/\text{C}.$$

0.3 point

(d) On the back face of the cube $d\vec{A} = (dA)(-\hat{k})$

$$\Phi = \int_{\text{back}} \vec{E} \cdot d\vec{A} = \int_{\text{back}} (E_x\hat{i} + E_y\hat{j}) \cdot (dA)(-\hat{k}) = 0. \quad 0.2 \text{ point}$$

(e) We now have to add the flux through all six faces. One can easily verify that the flux through the front face is zero, while that through the right face is the opposite of that through the left one, or $+16 \text{ N} \cdot \text{m}^2/\text{C}$. Thus the net flux through the cube is

$$\Phi = (-72 + 24 - 20 + 0 + 0 + 20) \text{ N} \cdot \text{m}^2/\text{C} = -48 \text{ N} \cdot \text{m}^2/\text{C}. \quad 0.5 \text{ point}$$

Note: The net charge enclosed by the cube can be got from Gauss' law:

$$q_{\text{enc}} = \epsilon_0 \Phi = 8.85 \times 10^{-12} \times (-48) = -42.5 \text{ nC}$$

The net charge enclosed by the cube is negative.

33. The charge on the metal plate, which is negative, exerts a force of repulsion on the electron and stops it. The electric field is given by $E = \sigma/\epsilon_0$, where σ is the surface charge density on the plate. The force on the electron is a constant force $F = -eE = -e\sigma/\epsilon_0$ and the work done by the electric force is:

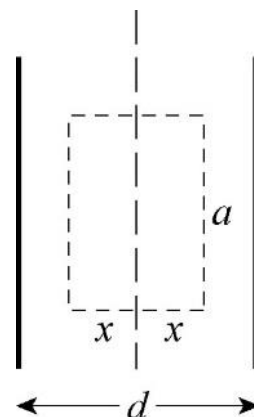
$$W = Fd = -\frac{e\sigma}{\epsilon_0} d$$

Apply the work-kinetic energy theorem, we have:

$$W = Fd = -\frac{e\sigma}{\epsilon_0} d = \Delta K = 0 - K_0$$

$$\Rightarrow d = \frac{\epsilon_0 K_0}{e\sigma} = \frac{(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(3.93 \times 10^{-17} \text{ J})}{(1.60 \times 10^{-19} \text{ C})(1.5 \times 10^{-6} \text{ C/m}^2)} = 1.45 \times 10^{-3} \text{ m}.$$

47. We use a Gaussian surface in the form of a box with rectangular sides. The cross section is shown with dashed lines in the diagram below. It is centered at the central plane of the slab, so the left and right faces are each a distance x from the central plane. We take the thickness of the rectangular solid to be a , the same as its length, so the left and right faces are squares.



The electric field is normal to the left and right faces and is uniform over them. Since $\rho = 1.89 \text{ fC/m}^3$ is positive, it points outward at both faces: toward the left at the left face and toward the right at the right face. Furthermore, the magnitude is the same at both faces. The electric flux through each of these faces is Ea^2 . The field is parallel to the other faces of the Gaussian surface and the flux through them is zero. The total flux through the Gaussian surface is $\Phi = 2Ea^2$. The volume enclosed by the Gaussian surface is $2a^2x$ and the charge contained within it is $q = 2a^2x\rho$. Gauss' law yields

$$2\varepsilon_0 Ea^2 = 2a^2x\rho.$$

We solve for the magnitude of the electric field: $E = \rho x / \varepsilon_0$.

(a) For $x = 0$, $E = 0$.

(b) For $x = 2.00 \text{ mm} = 2.00 \times 10^{-3} \text{ m}$,

$$E = \frac{\rho x}{\varepsilon_0} = \frac{(1.89 \times 10^{-15} \text{ C/m}^3)(2.00 \times 10^{-3} \text{ m})}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 0.427 \times 10^{-6} \text{ N/C}.$$

(c) For $x = d/2 = 4.70 \text{ mm}$

$$E = \frac{\rho x}{\varepsilon_0} = \frac{(1.89 \times 10^{-15} \text{ C/m}^3)(4.70 \times 10^{-3} \text{ m})}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 1.00 \times 10^{-6} \text{ N/C}.$$

(d) For $x = 26.0 \text{ mm}$, we take a Gaussian surface of the same shape and orientation, but with $x > d/2$, so the left and right faces are outside the slab. The total flux through the surface is again $\Phi = 2Ea^2$ but the charge enclosed is now $q = \rho a^2 d$. Gauss' law yields

$$2Ea^2 = \frac{\rho a^2 d}{\varepsilon_0} \Rightarrow E = \frac{\rho d}{2\varepsilon_0} = \frac{(1.89 \times 10^{-15} \text{ C/m}^3)(9.40 \times 10^{-3} \text{ m})}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 1.00 \times 10^{-6} \text{ N/C}.$$

49. As shown in figure, in the region I (left side of sheet 1), the net E field is:

$$E_I = -\frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0} - \frac{\sigma_3}{2\epsilon_0} = 0 \quad (1) \text{ 0.5 point}$$

In the region II (between sheets 1 and 2), the net field is:

$$E_{II} = \frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0} - \frac{\sigma_3}{2\epsilon_0} = 1 \times 10^5 \text{ N/C} \quad (2) \text{ 0.5 point}$$

In the region III (between sheets 2 and 3), the net field is:

$$E_{III} = \frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0} - \frac{\sigma_3}{2\epsilon_0} = 3 \times 10^5 \text{ N/C} \quad (3) \text{ 0.5 point}$$

From Eq. (1) & (3), we can get: $-\frac{\sigma_3}{2\epsilon_0} = \frac{3}{2} \times 10^5 \text{ N/C}$

From Eq. (2) & (3), we can get $\frac{\sigma_2}{2\epsilon_0} = 1 \times 10^5 \text{ N/C}$

So from above, we get: $\frac{\sigma_3}{\sigma_2} = -1.5 \quad \text{0.5 point}$

